

Containers in C++ STL

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Standard Template Library (STL) provides the built-in implementation of commonly used data structures known as containers. A container is a holder object that stores a collection of other objects (its elements). They are implemented as class templates, which allows great flexibility in the data types supported.

A container manages the storage space for its elements and provides member functions for easy access to useful operations.

Types of STL Containers

STL containers are divided into the following categories:

1. **Sequence Containers**
2. **Associative Containers**
3. **Unordered Containers**
4. **Container Adapters**

1. Sequence Containers

Sequence containers implement linear data structures in which the elements can be accessed sequentially. Following are the sequence containers in C++ STL:

Container Name	Description
<u>Array</u>	Container that wraps over fixed size static array.
<u>Vector</u>	Automatically resizable dynamic array.
<u>Deque</u>	Dynamic array of fixed-size arrays that allows fast insertions and deletions at both ends.

Container Name	Description
<u>List</u>	Implementation of Doubly Linked List data structure.
<u>Forward List</u>	Implementation of Singly Linked List data structure.

2. Associative Containers

Associative containers store data in some sorted order. It provides fast search, insert and delete in $O(\log n)$ time by using balanced trees like Red Black Trees.

Container Name	Description
<u>Set</u>	Collection of unique elements sorted on the basis of their values.
<u>Map</u>	Collection of key-value pairs sorted on the basis of the keys where no two pairs have same keys.
<u>Multiset</u>	Collection of elements sorted on the basis of their values but allows multiple copies of values.
<u>Multimap</u>	Collection of key-value pairs sorted on the basis of the keys where multiple pairs can have same keys.

3. Unordered Associative Containers

Unordered associative containers implement unsorted hashed data structures that can be quickly searched ($O(1)$ amortized, $O(n)$ worst-case complexity).

Container Name	Description
<u>Unordered Set</u>	Collection of unique elements hashed by their values.
<u>Unordered Map</u>	Collection of key-value pairs that are hashed by their keys where no two pairs have same keys.
<u>Unordered Multiset</u>	Collection of elements hashed by their values and allows multiple copies of values.
<u>Unordered Multimap</u>	Collection of key-value pairs that are hashed by their keys where multiple pairs can have same keys.

4. Container Adapters

Container adapters provide a different interface for other containers. They adapt the behavior of underlying containers to fit specific use cases.

Container Name	Description
<u>Stack</u>	Adapts a container to provide stack (LIFO) data structure.
<u>Queue</u>	Adapts a container to provide queue (FIFO) data structure.
<u>Priority Queue</u>	Adapts a container to provide heap data structure.

To master **C++ Standard Template Library (STL)** in the most efficient and effective way, do check out this [C++ STL Online Course](#) by GeeksforGeeks. The course covers the basics of C++ and in-depth explanations to all C++ STL containers, iterators, etc along with video explanations of a few problems. Also,

you'll learn to use STL inbuilt classes and functions in order to implement some of the complex data structures and perform operations on them conveniently.

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